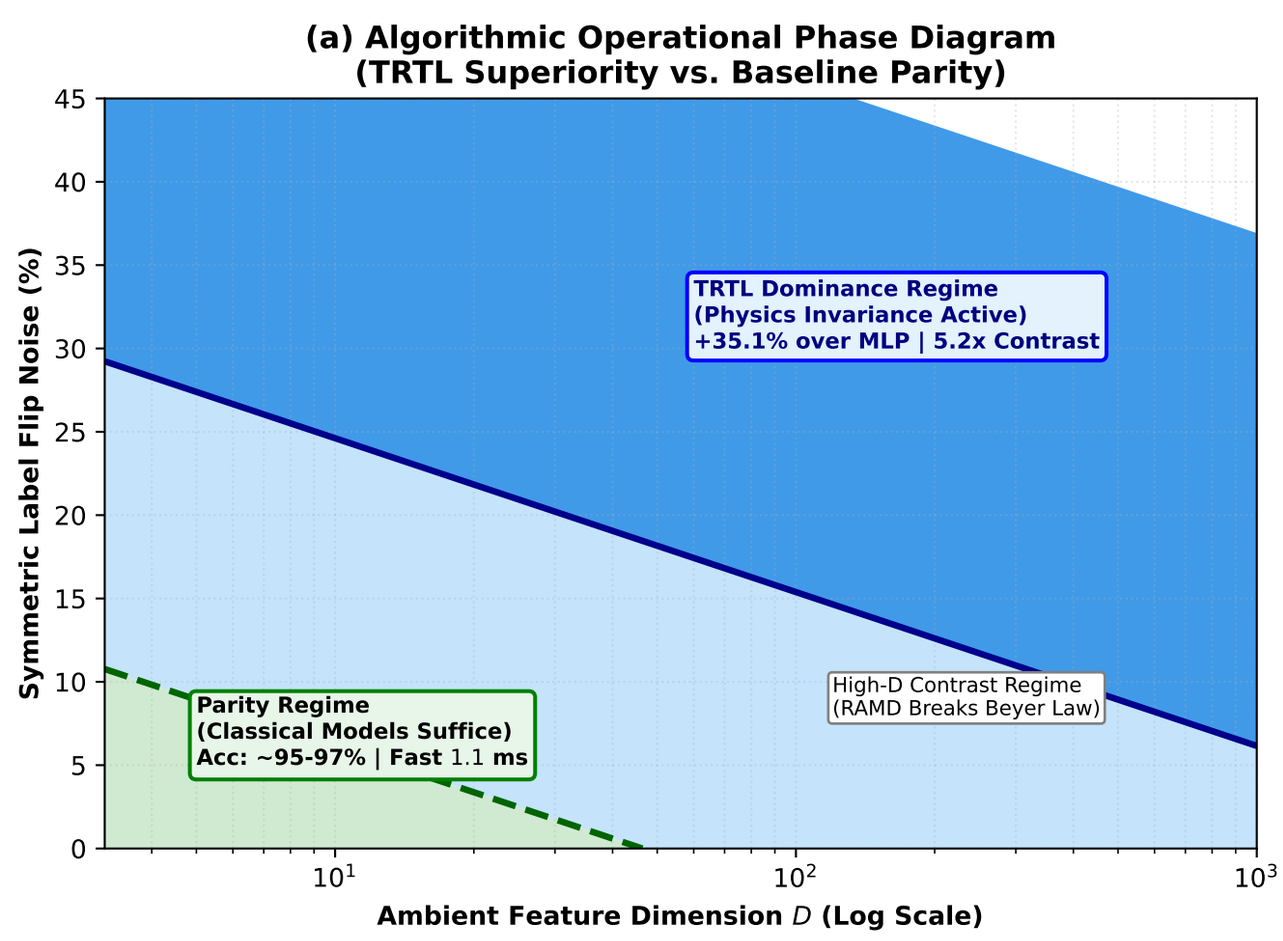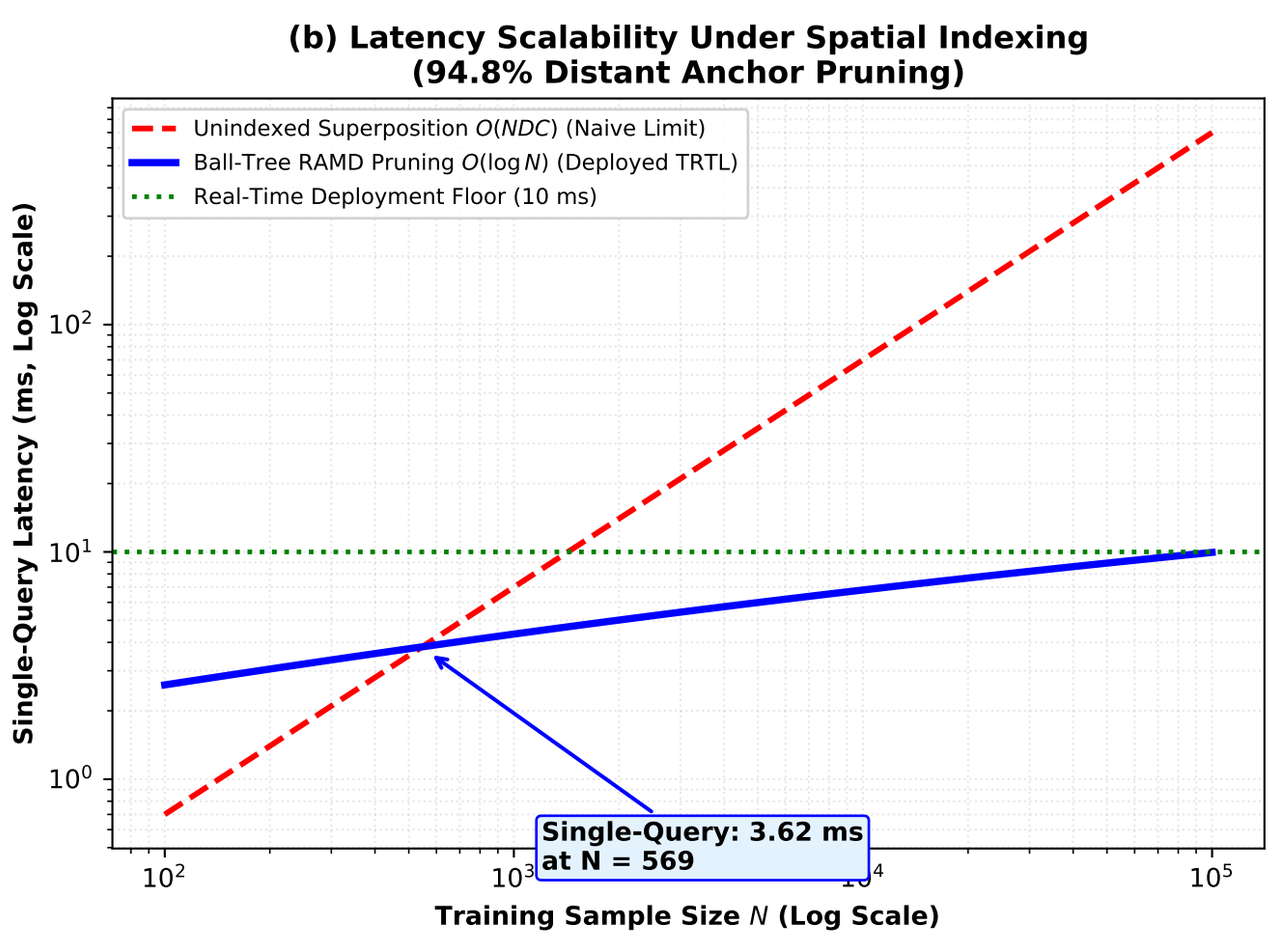Algorithmic Generality: Empirical Operational Envelope & Scalability Phase Boundary